Multi-objective Hybrid Fuzzified PSO and Fuzzy C-Means Algorithm for Clustering CDR Data

A. Jaya Mabel Rani and A. Pravin

***Abstract*—The growing field of mobile telecommunication becomes more and more competitive in the world. Therefore, the mobile operators are facing tremendous challenges. The customer requirements have increased in various levels of environments with huge amount of complexity of data. To solve this complexity of the problem a single-objective function may not offer suitable results of complex data. So this paper proposes a novel fuzzy clustering method based on Multi-Objective Hybrid Fuzzified particle swarm optimization and Fuzzy C-Means algorithm for the complex data. In this multi-objective hybrid clustering algorithm uses two objective functions: JFCM and the Xie-Beni index function. In the optimization level, FPSO find an optimal number of clusters by minimizing the value by Xie-Beni index. Based on the obtained number of clusters, FCM utilizes the optimal clustering centers and output the clustering results using JFCM objective function. Fuzzy C-Means (FCM) algorithm is the most liked and widely used clustering technique, because it is efficient and easy to execute. However, it suffers from the problem of initialization and is easily trapped in local optima. This clustering technique is used for predicting the social behavior and people's common interest, anti social thread by anti social communication. Then, based on business requirement, the service provider can predict which tower has to be changed, where to build a new tower and other such details based on the clustered data of customer's activities and behavior of different time period. The experimental results are shown for mobile call data records to obtain optimum and fast clustering solution.**

***Index Terms*—Call Data Records, Fuzzified Particle warm optimization, Fuzzy C-Means, Xie-Beni, Clustering.**

I. Introduction

OPTIMIZATION based clustering is widely studied in data mining, which are active real world problems in big data.Clustering is a grouping of data object, into set of disjoint collections of same objects[1]. The main aim of good clustering is to minimize intra-cluster distances [2] between data objects, and maximize inter-cluster distances [3] using an appropriate distance measure between data points. A distance measure [4] thus lies at the heart of data clustering. Most of the clustering techniques are in the form of unsupervised learning on which data points belong to which clusters [5] and this is the major difference between clustering (un supervised) and classification [6]. Clustering is also referred as automatic classification; however, this is inaccurate, since the clusters found are not known prior for processing. Recent research proved that the partitioned clustering techniques are well suited for clustering a large dimension of data sets; due to its low computational requirements [7]. The various partitioning clustering algorithm are K-means [8], K-harmonic Means [9], Fuzzy C-means algorithm [10], and etc., These algorithms are very simple, easy to apply in real time problems and less computation cost. In Fuzzy C-means clustering algorithm random initialization starts with predefined number of clusters [11]. The main disadvantage of the Fuzzy C-means algorithm is that, it is very sensitive to the random initialization of the cluster centroids may easily go for premature convergence [12] and also define the number of clusters. For global optimization solution, it is necessary to employee some large -scale optimal searching algorithm for generating this initial cluster centroids. Some of the intelligence techniques such as Genetic Algorithm (GA) [13], Particle Swarm Optimization (PSO) [14], Ant Colony Optimization [15] (ACO), Evolutionary Algorithms (EA) [16], Bacterial Foraging Optimization[17] (FSO), Biogeography Based Optimization (BBO) [18] etc., and various hybrid algorithms [19] have been successfully applied to the various global optimization clustering problems. The above near global optimization problems, required large number of iterations or high computation time. This paper proposed Hybrid FPSO and FCM with multi-objective function [20-23] for getting the advantages for both local and global optimization techniques.The rest of this paper is organized as below Section II describes the Related Work and finally Concludes the paper in Section III.

II. Related Work

In November 2017 Filippo Maria Bianchi, Antonello Rizzi, Alireza Sadeghian explained about call data records for identifying user behavior and regular habits using data mining based on LD-ABCD algorithm and also compared the output results of the knowledge discovery system by using PROCLUS [24]. This paper designed for better customer understanding habits of a telecommunication company. In march 2017 J. Kumaran Kumar, E. Karunakaran, K.M.Sabarivelananalysed Cluster for call data records by using Data Mining Technique to identify the user behaviour for telecommunication network from the collected input dataset [25].

Finally, a proposed clustering analysis analyze the group of user in the telecommunication based network, by using k-means clustering algorithm for recognizing the user behavior of the telecommunication network in the time duration of morning afternoon session and evening. In 2018 Call Detail Record (CDR) Analysis K-means Clustering using Tableau for 24 hours customer activities to create statistically-based grouping about similarities and performance of the groups compared with each other [26]. Using this clustering technique, they have analyzed about of making more traffic

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to the telecom network . This paper also analyze about call data records for telecommunication customer's behavior and activities and analyzed traffic location, and where to built new towers.

a) Fuzzy C-Means Algorithm

Fuzzy logic [27] is a mathematical logical model, which can be truth value i.e. it can have the value between 0 and 1, that is completely false (0) and completely true (1). The main aim of FCM algorithm is to minimize the objective function denoted as J_{FCM} by Euclidean distance using the following equation (1);

$$J_{FCM} = J(U,V) = \sum_{i=1}^{N}\sum_{j=1}^{C}(U_{ij})^{m}\left\|D_i - C_j\right\|^2,$$

where m≥1 (1)

Here *m* is fuzzification degree; that is any real number greater than 1, $U=[U_{ij}]_{i=1\ldots N,\ j=1\ldots C}$ is an N×C membership matrix;denotes the membership degree of D_i in j^{th} cluster. $C=[C_1\ C_2\ \ldots C_j]$is an $I\times J$ matrix, whose columns corresponding to cluster centres.Then D_i isi^{th} dimension of the measured data, and $C_j$is the jth dimension of the cluster center. The cluster center is updated by the following equation (2);

$$C_j = \frac{\sum_{i=1}^{N} U_{ij}^{m}.D_i}{\sum_{i=1}^{N} U_{ij}^{m}} \quad (2)$$

The fuzzy membership partition matrix [28] values are updated by the equation (3);

$$U_{ij} = \frac{1}{\sum_{k=1}^{C}\left(\frac{\left\|D_i - C_j\right\|}{\left\|Di - C_k\right\|}\right)^{2/m-1}} \quad (3)$$

The steps for Fuzzy C-Means algorithm is given below.
Let us consider

$D=\{D_1, D_2, D_3, \ldots D_n\}$ be the set of data points and $C=\{C_1, C_2, C_3, \ldots C_c\}$ are the set of cluster centers.

Step 1: Randomly select cluster centers.
Step2: Compute the fuzzy membership, U_{ij} using the equation (3).
Step 3: Then find the fuzzy cluster center C_j using the equation (2).
Step4: Repeat step 2 and 3 until the number of iterations

$$\left\|(U_{ij})^{k+1} - (U_{ij})^{k}\right\| < \varepsilon \quad (4)$$

Where, *k* is the iteration step, ε is the termination condition between 0 and 1. The FCM algorithm's iteration stops when the maximum change in the values of partition matrix is less than ε; defined as 0.00015. This algorithm is more advantage than K-Means; still it has some disadvantage to compare with global optimization techniques. The main weakness of this FCM is sensitiveness of the initialization to cluster centered, need to determine membership cut -off value and need to define number of clusters, which is not a deterministic algorithm [29].

b) Particle Swarm Optimization Algorithm

In recent clustering analysis, Particle Swarm Optimization (PSO) is being widely executed due to its effectiveness when compared with traditional clustering algorithms like *K*-Means and its variants and other clustering approaches. The PSO technique produces superior clustering results [30] with reference to the intra and inter-cluster distances measure. Eberhart and Kennedy in 1995 [31] designed PSO, and was motivated by the social activities of a flocking of birds. In PSO technique [32] the birds in a flock are emblematically represented as particles. The values c_1 and c_2 called control weight used to balance the *pbest* and *gbes*t in deciding particle's next movement velocity. The mathematical updating of position and velocity in the multi dimensional problem space is according to the following equations (5, 6)

$$v_i^{it+1} = kc\ *\left(\omega * v_i^{it} + c_1\,.\,rand_1\left(pbest_i^{it} - d_i^{it}\right) + c_2.rand_2\left(gbest^{it} - d_i^{it}\right)\right) \quad (5)$$

$$d_i^{it+1} = d_i^{it} + v_i^{it+1} \quad (6)$$

Where *kc* is the constriction factor.

$$kc = \frac{2}{\left|2-c-\sqrt{c^2-4c}\right|} \quad (7)$$

Where $c = c_1 + c_2$ and $c > 4$. c_1 and c_2 are acceleration constants, $d_i^{it}$and v_i^{it} are the current position and velocity of the ith particle. The random values $rand_1, rand_2$are in the range of (0, 1); The ω is calculated by the following equation (8)

$$\omega = \omega_{max} - \frac{\omega_{max} - \omega_{min}}{it_{max}} \times it \quad (8)$$

The best performance fitness value is calculated for each iteration, based on equation (9).

$$pbest_i^{(it+1)} = \begin{cases} pbest_i^{it} & f\left(d_i(it+1)\right) \le f\left(d_i(it)\right) \\ d_i(it+1) & f\left(d_i(it+1)\right) > f(d_i(it)) \end{cases} \quad (9)$$

Here f *d*enotes the performance fitness function, $pbest_i^{it}$ stands for the best fitness values and *it* denotes the generation step.

In Fuzzy logic, the entire data set range lies between 0 and 1.In FPSO the performance fitness function consider as a Xie Beni Index function [33].

XIE BENI Index

Xie Beni index value takes the minimum value for the objective function for its optimal value Xie –Beni index is defined in the following equation (10),

$$U_{XB}(U,V;D) = \frac{\sum_{i=1}^{c}\sum_{j=1}^{n}(u_{ij})^{m}\|D_i - C_j\|^2}{N \min_{i\neq k}\|C_j - C_k\|^2} \quad (10)$$

The steps involved in the FPSO is given as follows,

(i) The fuzzy logic inputs and output are decided and their feasible ranges are declared. They are as follows:

$$Input\ 1 = f/f_{max} \quad (11)$$

$$Input\ 2 = Max\ \{(X_i^{max\ it} - X_i^{it\ iter})\ (X_i^{\ it\ iter} - X_i^{min\ it}) \quad (12)$$

The Input 1 is the first important input based on current and maximum fitness function and Input 2 determine the maximum search distance vector.

(ii) Fuzzification of inputs and output using triangular membership function with seven fuzzy linguistic sets have been used for each of the inputs and output as shown in Fig. 1. In this fuzzy membership function X axis indicates the crisp values from 0.0 to 2.2 and 0 to 1.

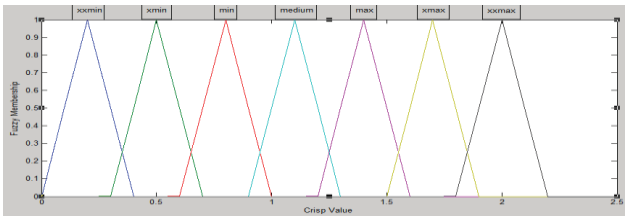


Fig. 1. Fuzzy Membership Function

(iii) The fuzzy rule based formulation for all combinations of fuzzy inputs based on their ranges is given in the following Table I. In Table I, seven fuzzy linguistic sets such as XXMin, XMin, Min, Medium, Max, XMax, XXMax and its corresponding fuzzy rules based on all combination of fuzzy input. In Table II given the clustering performace based on purity and CPU execution time.

(iv) Defuzzification of fuzzy output is shown by using Centroid method.

$$C = \sum_{i=1}^{7} D_i\ Y_i / \sum_{i=1}^{7} Y_i \quad (13)$$

TABLE I
COMBINATION OF FUZZY INPUTS BASED ON FUZZY RULE

I/P 1 / I/P 2	XXMin	XMin	Min	Medium	Max	XMax	XXmax
XXMin	XXMin	XXMin	XMin	XMin	Min	Min	Medium
XMin	XXMin	XMin	XMin	Min	Min	Medium	Medium
Min	XMin	XMin	Min	Min	Medium	Medium	Max
Medium	XMin	Min	Min	Medium	Medium	Max	Max
Max	Min	Min	Medium	Medium	Max	Max	XMax
	Min	XMax	Medium	Max	Max	XMax	XMax
XXMax	Medium	Medium	Max	Max	XMax	XMax	XXMax

c) Multi-Objective Fuzzified Particle SwarmOptimization And Fuzzy C-Means Algorithm.

The proposed Multi-objective FPSO and FCM algorithm is designed in two step process.

(i) Find Optimal number of clusters using Xie-Beni index function.
(ii) Obtain optimal clustering solution using FCM function.

The steps for Multi-Objective FPSO and FCM algorithm, is designed as follows.

Step 1: Generate the initial population $P=P_1, P_2, \ldots\ldots P_n$. Where n is the population size.
Step 2: Then select the cluster centers $C= C_1, C_2, \ldots. C_k$ based on fuzzy adaptive rules given in the section 4. K is the number of clusters.
Step 3: Calculate fitness function using Xie_Beni index function using equation (13).

TABLE II
CLUSTERING PERFORMANCE BASED ON PURITY, CPU EXECUTION TIME

Clustering Algorithm	Number of clusters	Purity	CPU time taken in Seconds
FCM	4	0.72	.3 Seconds
FPSO+FCM with SOF	7	0.86	2.25 Seconds
FPSO+FCM with MOF	7	0.8702	2.02 Seconds

Step 4: Update velocity and position of each particles by gbest and pbest.

Step 5: Update the cluster centers by a new updated values and fuzzy based adaptive rules.

Step 6: Check the stopping criteria. If no go to step 3, else go to Step 7 to find optimal clustering solution by using current number of cluster centered.

Step 7: Consider the current number of cluster centers as input to fuzzy C-Means algorithm (ie initial cluster center), to find optimal clustering solution using J_{FCM} objective function.

Step 8: Update fuzzy membership values for each candidates of the solution.

Step 9: Calculate fitness function using the objective function of J_{FCM} by using equation (1) for every particles in the swarm.

Step 10: After calculation of objective function cluster centers are updated using equation (2) and (3).

Step 11: Then select best fitness function.

Step 12: Check Stopping condition. If yes go to end, Else go to step 8**.**

Step 13: End

In this data set contains 10 various attributes such as MSIDN, IMSI, IMEI, plan of calling, call type, corresponding type, corresponding ISDN, call duration, time, Date with 728 instances and tested with Fuzzy C-Means, FPSO+FCM with single objective function and FPSO+FCM with multi objective function. Nowadays mobile phones are very good excellent tools for studying of human mobility analysis [34-36] Call Detail Records collected all data from all mobile phones, smarts, dumb, etc within few seconds after usage, and provide complete records of mobile phone usage. This algorithm analyses the groups of users, members of these groups of mobile phone communication, types of calling, etc. Each group has a separate calling signature. Each person of the phone number involved voice call or SMS message. Each call data information also contains the starting time of the call, call duration of the event and the location of the cell tower associated for the event. The above Table II shows the clustering performance based on purity, CPU execution time in seconds. In this analysis FCM taken less computation cost than FPSO+FCM algorithm. But the cluster purity of multi-objective hybrid FPSO+FCM is higher than FCM.

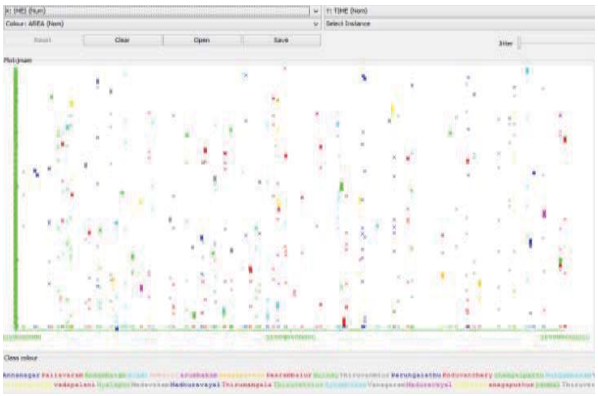


Fig. 2. Clustering Visualization Only Based On Area.

Fig. 2 shows the Clustering visualization only based on area. Each colour shows the different area of the mobile call.

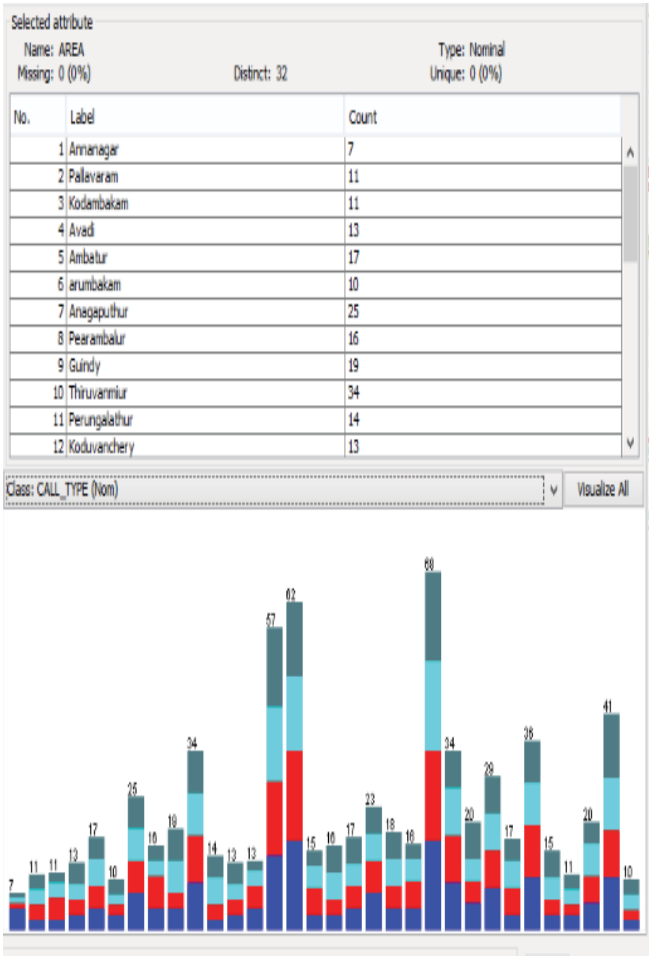


Fig. 3. Overall Clustering Visualization For Mobile Call Data Records

The Fig. 3 shows the overall clustering visualization for CDR data in 2D form.

III. Conclusion

Thus the Multi-objective hybrid FPSO+FCM methodology is examined for Mobile call data records. It is found that, this hybrid clustering effectively tackled CDR clustering problems. FPSO+FCM algorithm combines the ability of fast convergence of FCM and global optimization of FPSO, to avoid the premature convergence. Due to its simplicity and efficiency and multi-objective function of this algorithm we can use for large search spaces for optimal clustering solutions. The output of FPSO used as the initial number of clusters for Fuzzy C-Means, which quickly locates the optima with better fitness value than with random initial number of clusters. However, the proposed FPSO models handle the stagnation of data, avoid the premature convergence of the particles and also avoid the initialization of FCM algorithm. This algorithm was also tested with real time mobile data records.

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